AI FUEL EVENT DETECTION

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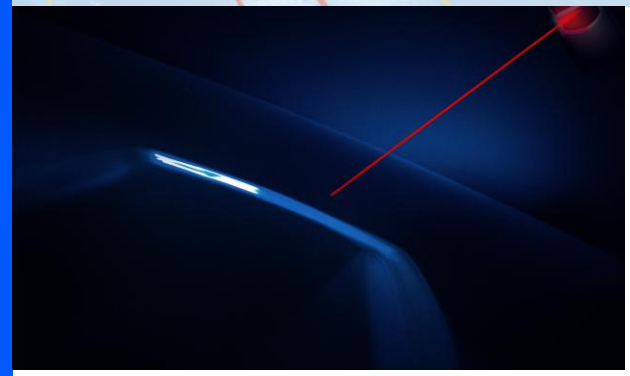


SICK Sensor Intelligence

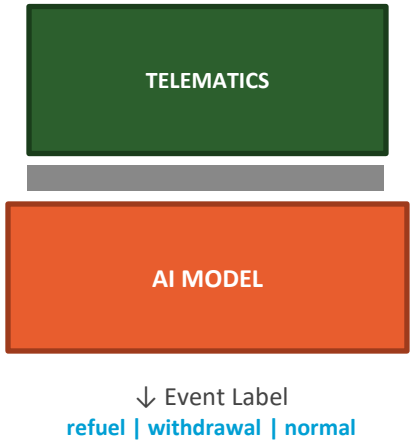
REAL WORLD CONTEXT

The Setup

Each truck is equipped with a **fuel probe** that constantly sends fuel level readings (in litres) **every 2 seconds**. These readings are processed through our pipeline and stored in a database together with **obj_id, time** and an event **flag**. The **contact** and **speed** values are fetched separately via a special query and joined to each record.

The Challenge

Given this stream of data, students must build a model that can **correctly identify fuel events in real time** — something the existing rule-based algorithm has failed to do reliably. The model must distinguish between **refuelling, withdrawal**, and **normal consumption** — using only the sequence of sensor readings available at that moment.



BUSINESS CASE

- **Existing Rule-Based Algorithm**

A live fuel algorithm is currently in production, classifying fuel events in real time: refuelling, withdrawal, and oscillation events for individual vehicle tanks.

- **Problem: Detection Failures**

The existing algorithm has **failed to detect obvious refuelling and withdrawal events**. An AI-powered solution is needed to correctly classify fuel events in real time: **refuelling, withdrawal, normal consumption**, and oscillations (bonus).

- **Training Dataset**

Each sample is a **sequence of ~60 frames** (adjustable), with frame intervals of **10s, 20s, 30s, or 1 min** (adjustable). Features per frame: **time, obj_id, fuel_lvl, speed, contact, label**.

WHY RULE BASED ALGORITHMS FAIL IN FUEL EVENT DETECTION

1. Rule based detection (today)

Fleet management systems detect fuel events using simple rules such as:

- Fuel increase > 20 l → refuelling
- Fuel decrease > 20 l → fuel withdrawal
- Vehicle speed = 0
- Engine = off

2. Problem in real world data

Vehicle telemetry data is often noisy and unpredictable:

- Drivers may refuel while the engine is running
- Gps may report small movement when the vehicle is parked
- Fuel sensors oscillate due to tank movement in the tank
- Fuel theft may happen slowly over time

3. Consequences

Rule based systems can produce:

- Missed refuelling events
- Missed fuel theft events
- False alarms caused by sensor oscillations

4. Proposed solution

Use a machine learning model that analyzes sequences of telemetry data and classifies:

- Refuelling
- Fuel withdrawal
- Normal consumption
- Sensor oscillation

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BUSINESS CASE - SCENARIO 1
STANDARD REFUELLING AT A FUEL STATION

Situation

A truck arrives at a fuel station and refuels before continuing the trip.

Telemetry pattern

Fuel level increases rapidly within a short time window.

Example:

120 L → 150 L → 190 L → 220 L

Vehicle state:

- speed = 0 km/h
- contact = OFF or ON

Event that must be detected

Refuelling

Business importance

- Fleet managers must verify that fuel purchases match the recorded refuelling events.

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BUSINESS CASE - SCENARIO 2
REFUELLING WHILE ENGINE IS RUNNING

Situation

Some drivers leave the engine running while refuelling, especially in refrigerated trucks or construction vehicles.

Telemetry pattern

Fuel level increases quickly.

Example:

90 L → 120 L → 150 L

Vehicle state:

- speed = 0 km/h
- contact = ON

Event that must be detected

Refuelling

Business importance

- Fuel reports must remain accurate regardless of driver behaviour.

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BUSINESS CASE - SCENARIO 3
FUEL THEFT WHILE VEHICLE IS PARKED

Situation

Fuel is stolen from a vehicle parked overnight at a parking lot or rest area.

Telemetry pattern

Fuel level suddenly decreases without vehicle movement.

Example:

200 L → 180 L → 160 L → 150 L

Vehicle state:

- speed = 0 km/h
- contact = OFF

Event that must be detected

Fuel withdrawal

Business importance

- Fleet operators need to detect fuel theft and prevent financial losses.

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BUSINESS CASE - SCENARIO 4
FUEL WITHDRAWAL WHILE ENGINE IS RUNNING

Situation

Fuel is siphoned from a vehicle while it is stopped but the engine remains running (for example during loading or waiting periods).

Telemetry pattern

Fuel level decreases rapidly.

Example:

180 L → 170 L → 155 L → 140 L

Vehicle state:

- speed = 0 km/h
- contact = ON

Event that must be detected

Fuel withdrawal

Business importance

- The system must detect abnormal fuel loss even when the vehicle engine is running.

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BUSINESS CASE - SCENARIO 5
REFUELLING AT A COMPANY DEPOT

Situation

Vehicles are refuelled at a private company fuel station rather than a public gas station.

Telemetry pattern

Fuel level increases within a few minutes.

Example:

70 L → 100 L → 130 L

Vehicle state:

- speed = 0 km/h
- contact = OFF

Event that must be detected

Refuelling

Business importance

- Allows monitoring of company fuel inventory and internal fuel consumption.

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BUSINESS CASE - SCENARIO 6
NORMAL FUEL CONSUMPTION

Situation

A vehicle is driving normally and consuming fuel during the trip.

Telemetry pattern

Fuel level gradually decreases over time.

Example:

200 L → 198 L → 196 L → 194 L → 192 L

Vehicle state:

- speed = 70–90 km/h
- contact = ON

Event that must be detected

Normal fuel consumption

Business importance

- The system must distinguish normal fuel usage from fuel theft to avoid false alarms

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BUSINESS CASE - SCENARIO 7
FUEL SENSOR OSCILLATION

Situation

Fuel inside the tank moves while the vehicle is driving on uneven roads, causing unstable sensor readings.

Telemetry pattern

Fuel level fluctuates up and down.

Example:

100 L → 92 L → 104 L → 95 L → 101 L

Vehicle state:

- speed = 40–60 km/h
- contact = ON

Event that must be detected

Fuel sensor oscillation (not a real fuel event)

Business importance

- Incorrect classification could generate false refuelling or theft alerts.

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DATASET EXAMPLE

Each sequence = 60 frames (1 row per frame).
Label column is the **target flag**: 0 = normal consumption, 1 = refuel, 2 = withdrawal. Two example sequences shown below.

seq_id	time	obj_id	fuel_lvl	speed	contact	frame	label
SEQ_001	2024-03-01 08:00	VEH_042	45.2	0	0	1	**1 (refuel)**
SEQ_001	2024-03-01 08:01	VEH_042	62.8	0	0	2	**1 (refuel)**
... (60 rows)							
SEQ_002	2024-03-01 10:15	VEH_117	78.0	72	1	1	**0 (normal)**
SEQ_002	2024-03-01 10:16	VEH_117	77.4	68	1	2	**0 (normal)**
... (60 rows)							

GOAL

- Detect **fuel-related events** from vehicle telemetry time-series data:
  - **Refuel** (fuel increase)
  - **Fuel withdrawal / theft** (fuel loss)
  - **Normal consumption** (no event)
- This is framed as a **sequence labeling problem**, where:
  - **Input** = time-ordered sensor readings
  - **Output** = label per timestep indicating fuel event state

INPUT ATTRIBUTES (FEATURES)

- Each timestep t in the sequence contains:
- **Fuel LvL**
  - Description: Fuel level in the tank (liters).
  - Role in detection:
    - Primary signal for identifying fuel changes
    - Sudden increase → possible refuel
    - Sudden decrease → possible withdrawal
- **Important considerations:**
  - Sensor noise and quantization
  - Lag or smoothing in fuel sensors
  - Calibration differences per vehicle

SPEED

Description: Vehicle speed (km/h).
Role in detection:

- Helps distinguish **normal consumption** from suspicious events
- Fuel drop while speed = 0 → high withdrawal probability
- Fuel drop while driving → normal usage

Derived features (recommended):

- is_moving = speed > threshold
- Rolling average speed
- Stop duration counters

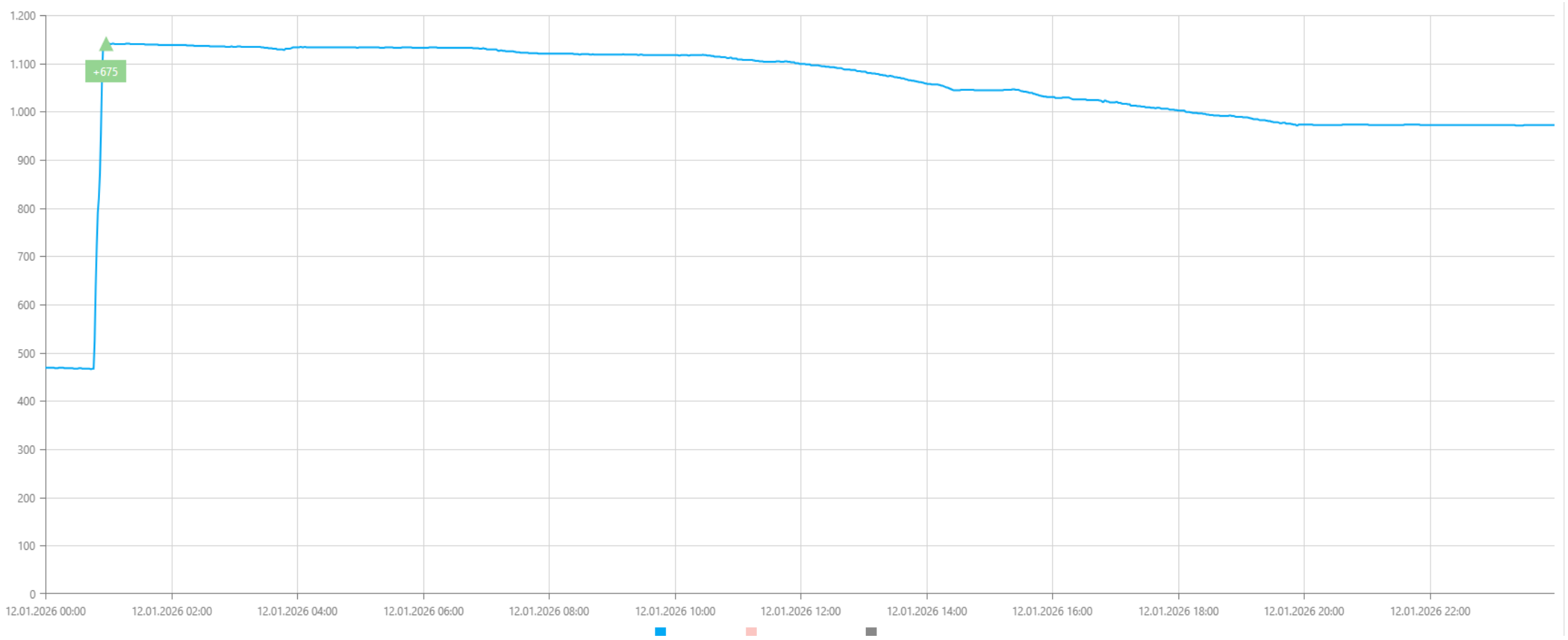
CONTACT

Description: Ignition or engine contact state (0 = off, 1 = on).

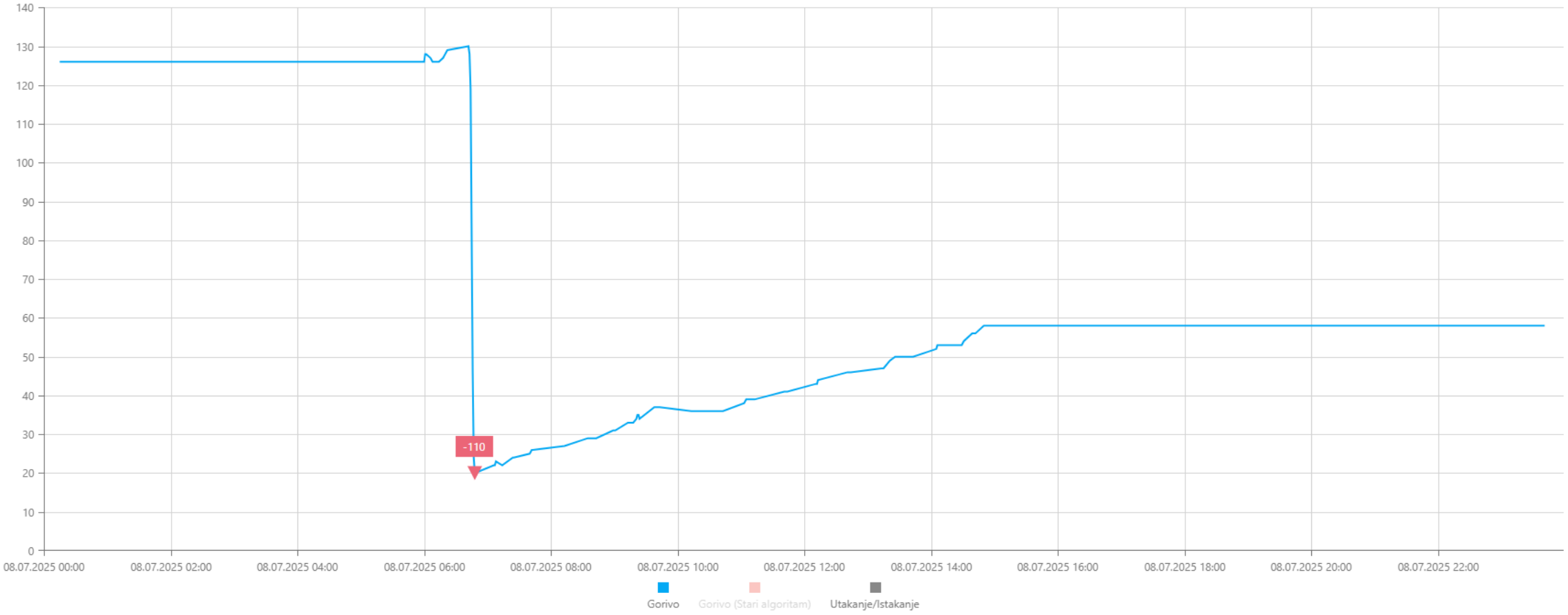
Role in detection:

- Refueling usually occurs when:
  - contact = off
  - speed = 0
- Fuel withdrawal often happens with:
  - contact = off
  - speed = 0
- Sudden fuel change with contact = on → possible sensor anomaly

REAL WORLD CASE : REFUEL FUEL EVENT



REAL WORLD CASE: WITHDRAWAL FUEL EVENT



THANK YOU.

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Q&A

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